

Babäk Firoozi Fooladi

Location: Espoo, Uusimaa, Finland
LinkedIn | GitHub | StackOverFlow
Email: b.firoozi.f@gmail.com

Data Engineer | Analytics Engineer

Data Engineer with a PhD background (Aalto University) and hands-on experience building **production data pipelines** and **reproducible analysis workflows** in **Python**, **R** and **SQL**. Proficient in biostatistics, survival models and econometrics. Architected a partitioned **data lake** with **DuckDB** and full documentation for a research team, built automated **Azure Functions** pipelines for acquiring and processing data from API endpoints, and led a **Pandas-to-Polars** migration with new **test coverage** for a 60% overall speedup. Keen to bring a focus on **data quality**, **lineage**, and **reproducibility** in collaboration with professionals and researchers.

Core Competencies & Technical Skills

- **Position related skills:** DBT, SQL, Python, Pandas, Polars, Git, R, quantitative data, biostatistics
- **Data Analytics & BI:** Power BI, Dashboard Development, Data Visualization, Quantitative Research, Business Intelligence, Technical Documentation
- **Programming & Core Languages:** Python, R, SQL, Julia, STATA, Git
- **Data Engineering & Databases:** PostgreSQL, PostGIS, DuckDB, Pandas, Polars, GeoParquet, SQLite, MongoDB, REST APIs, Web Scraping, ETL/ELT Pipelines, Azure Functions, GCP, Azure
- **Spatial Data & GIS Analytics:** GeoPandas, Shapely, ArcGIS / ArcGIS Online, ESRI CityEngine, Spatial Econometrics, Cartography, Network Analysis
- **Data Science & ML:** Machine Learning, Econometric Modeling, Time-Series Forecasting (VAR, Co-integration, Univariate/Multivariate), OLS / IV-OLS, Panel Data, Survival Analysis, RAG, Agentic AI, LLMs/NLP

Professional Experience

DATA ANALYTICS & ENGINEERING Jan 2025 -- Present
Self-Employed | Consultant *Project-Based -- Espoo, Finland*

- Deliver end-to-end **data engineering**, spatial analytics, and business intelligence solutions tailored to client objectives and scalable growth.
- Designing data processing pipeline in **DBT** connected to ducklake and DuckDB, **SQL** and **Python**
- **Backend Architecture & Spatial Engineering (Apex-Heat):** Architected automated Python scripts using GeoPandas, Shapely, GeoParquet, and PostGIS to acquire, clean, geometry-correct, and model national spatial building datasets across Finland.
- **API & Data Pipeline Automation:** Designed and implemented API-driven data pipelines for automated spatial operations, evaluating data models for EU-wide scaling using domain expertise in real estate economics and zoning laws.
- **Pipeline Optimization & Tech Debt Reduction (Apex-Heat):** Spearheaded core codebase migration from Pandas to Polars, refactoring design patterns and introducing comprehensive test coverage to achieve an **86% performance improvement** on targeted calculations and a **60% overall application speedup**.

Ph.D. Candidate Sep 2020 -- June 2026
Aalto University *Full-time -- Espoo, Finland*

- Conduct quantitative research at the intersection of urban economics and real estate, applying econometrics and machine learning to housing market dynamics and spatial development.
- Engineered automated web scraping pipelines using Azure Functions and REST/ArcGIS endpoints to harvest, geocode, and store real-world property transaction data (MML, Ryhti, SYKE, KVKL).
- Modeled complex urban dynamics using time-series analysis, hedonic regressions, and time-to-event statistical modeling.
- Applied NLP and RAG architectures to process PDF/HTML land-use planning documents for structured feature extraction.

Research Assistant

Jun 2019 -- Sep 2019

Aalto University – Full-time – Espoo, Finland

- Collaborated with **MIT CityLab** on Agent-Based Modeling (ABM) to simulate urban mobility and parking pressure dynamics across campus infrastructure using the GAMA Platform.

Data Analytics Engineer

Sep 2020 -- Dec 2024

Qissa kaupunkisuunnitteluanalytiikka Oy -- Startup

Part-time -- Espoo, Finland

- Designed and maintained end-to-end spatial data pipelines ingesting REST API data, performing geocoding and automated storage for urban planning analytics.
- Developed a spatial analytics dashboard with a custom network analysis backend, integrating HSL, HERE, and HSY APIs for real-time transportation routing and network evaluation.
- Processed multi-national census datasets (US, Finland, Sweden, Denmark) to calculate commuting pattern carbon emissions.

Planning Assistant

Nov 2018 -- Jun 2021

WSP Finland Oy

Part-time -- Helsinki, Finland

- Performed spatial data analysis and pedestrian network modeling to forecast route choices and post-development pedestrian traffic for municipal clients.
- Built interactive 3D city models using ESRI CityEngine and configured ArcGIS Online platforms for maritime spatial planning projects.
- Produced BI reports, spatial maps, and visualizations to communicate quantitative findings to multidisciplinary teams and executive stakeholders.

Education

Aalto University

Espoo, Finland

Doctor of Philosophy in Technology, Real Estate Economics

Sep 2020 - Expected Dec 2026

Aalto University

Espoo, Finland

MSc. (Tech.) Urban Studies and Planning in Real Estate Economic

Sep 2018 - Jul 2020

GPA: 4.1/5, Graduated with honours

Languages

- English (Fluent :C2)
- Finnish (beginner :A1-A2)

Publications:

- Effects of zoning on property values and impact assessment of changes in land use policy measures: [Source](#)

Certificates:

- Oxford Machine Learning Summer School, OxML 2023
- Urban Economic Association, UEA 2020 Lectures on Urban Economics

Community & Leadership

- **Speaker:** "What is Survival Analysis?". PyData Helsinki, *12th August 2026, Visma HQ, Helsinki, Finland*
- **Speaker:** "Balancing financial constraints and public objectives: Assessing the impact of public land ownership on housing development duration". World Planning School Congress, *17th-21st July 2026, Espoo, Finland*
- **Contributor:** Helsinki Data Week volunteer (*October 2024 & 2025*), *Helsinki, Finland*
- **Speaker:** Spatial data analysis, spatial data engineering. DataTribe collective meetup, *2023, Helsinki, Finland*
- **Speaker:** "The Effect of Land Development Policies on Implementation of Housing Plans". 28th Annual European Real Estate Society conference, *22nd-25th June 2022, Milan, Italy*
- **Speaker:** "The Impact of Public Land Development on Implementation of Housing Plans". AREUEA International Conference, *19th-22nd July, 2023, Cambridge, UK*
- **Speaker:** "Time limit: is it an effective way to encourage housing construction?". RYTS Science Slam, *25th of January 2023, Helsinki, Finland*
- DataTribe collective moderator
- **Guest Lecturer:** Science and Breakfast Staff Meeting, Introduction to Zotero, how to migrate from Mendeley. *October 2023, Espoo, Finland.*
- **Guest Lecturer:** Guest lecturer in Spatial Planning and Traffic engineering (SPT) program, Aalto university. Data analysis and Visualization using PPGIS. *March 2023, Espoo, Finland.*
- **Guest Lecturer:** Guest lecturer in Urban Studies and Planning (USP) program, Aalto university. How to use GIS in Urban studies and planning. *January 2023, Espoo, Finland.*
- **Guest Lecturer:** Tutor in Urban Studio Challenge 1 (USP) course, Data analysis, GIS, planning and data visualization assistant. *September 2022, Espoo, Finland.*
- **Guest Lecturer:** Guest lecturer in Sustainable Built Environment (SBE) course, Aalto university. GIS, Spatial Thinking and their use cases in sustainability and urban resilience. *November 2021, Espoo, Finland.*
- **Guest Lecturer:** Guest lecturer in Sustainable Built Environment (SBE) course, Aalto university. GIS and its use cases in sustainability and urban resilience. *December 2020, Espoo, Finland.*
- **Guest Lecturer:** Guest lecturer in Digital urban course at Aalto university. Teaching about data analysis and its uses in built environment, types of data and available information. *March 2020, Espoo, Finland.*